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Scalable adaptive assessment framework for collaborative eLearning: a service-oriented AI approach

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Abstract

Traditional eLearning assessments frequently rely on static, non-adaptive methods that provide limited personalized feedback and face significant scalability constraints. The Service-Oriented Framework for Intelligent Assessment (SOFIA) integrates Service-Oriented Architecture (SOA) principles with AI-driven adaptive feedback mechanisms to deliver real-time, personalized assessments at scale. Its modular, microservices-based architecture enables seamless integration with Learning Management Systems (LMS) and supports a wide range of educational contexts. The framework processes real-time learner interaction data using priority-based task optimization algorithms and generates iterative feedback informed by performance metrics. An experimental study involving 250 participants in an online Python programming course demonstrated its effectiveness. SOFIA achieved 92% accuracy in identifying learning gaps, outperforming static assessments (65%) and rule-based adaptive systems (78%). Learner engagement increased by 40%, and post-test retention scores improved to 85% compared with 65% in the control group. Scalability tests confirmed the framework's capacity to efficiently support more than 500 concurrent users. SOFIA addresses key limitations of conventional eLearning assessments by enhancing engagement, performance, and retention. It provides a scalable and adaptive solution that integrates seamlessly into existing LMS platforms, equipping educators with automated personalized assessment and feedback tools. While the study focused on Python programming, future research should examine its applicability across broader disciplines and pedagogical models. This work establishes a strong foundation for next-generation intelligent educational technologies by combining adaptive assessment with service-oriented design principles.

Keywords: Collaborative learning, Adaptive eLearning, Service-oriented computing, Real-time feedback, Predictive analytics, Scalable educational systems

Introduction

Collaborative eLearning has become a core pedagogical approach, supported by evidence that group interaction fosters engagement, self-regulation, and knowledge co-construction (Azevedo & Gašević, 2019; Dillenbourg, 1999; Järvelä et al., 2023; Johnson & Johnson, 2009). Learners benefit from shared problem-solving, negotiation

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of meaning, and distributed cognitive processes that deepen understanding. Assessing these dynamic, socially regulated processes remains difficult in practice, especially when courses scale and interactions evolve rapidly over time.

Traditional assessment approaches are typically static and uniform, oriented toward linear, individual activity. Limited capacity to adapt to evolving learner interactions and contextual shifts reduces the value of feedback, which often arrives late and lacks situational relevance (Black & Wiliam, 2009; Pellegrino et al., 2001; Sadler, 1989). These limitations are particularly pronounced in collaborative settings, where learning is co-regulated and shaped by ongoing peer interactions (Hadwin et al., 2018). Socially shared regulation of learning offers a useful lens, highlighting the need for assessments that are adaptive and sensitive to group-level processes.

Recent work in AI-enhanced assessment points to feasible pathways for real-time analytics, context-aware feedback, and personalization at scale (Hadyaoui & Cheniti-Belcadhi, 2025; Zawacki-Richter et al., 2019; Zheng et al., 2022). Generative AI introduces further opportunities for context-sensitive prompting and feedback phrasing, with potential to reduce instructor workload and support equitable distribution of feedback within teams (Baidoo-Anu & Owusu Ansah, 2023; Khosravi et al., 2025). Despite these advances, many systems remain difficult to deploy institution-wide due to architectural rigidity and limited interoperability.

Research gap and study objectives

A clear gap persists at the intersection of service-oriented architecture (SOA) and AI-driven adaptive assessment for collaborative eLearning. Existing solutions primarily focus on individual personalization or isolated feedback mechanisms, offering limited support for group-level dynamics, task orchestration, and real-time responsiveness at LMS scale (Järvelä et al., 2023; Sharma et al., 2024). SOA provides modularity, interoperability, and scalability that are well-suited to institutional contexts (Choi et al., 2025; Dragoni et al., 2017; Papazoglou et al., 2018). However, few frameworks unify real-time analytics, group-aware task allocation, and seamless LMS integration within a single adaptive assessment system. As a result, institutions face fragmented feedback processes, instructors experience increased coordination overhead, and learners receive uneven support in collaborative settings.

This study introduces the Service-Oriented Framework for Intelligent Assessment (SOFIA) to address this gap. SOFIA is designed to deliver real-time, scalable, and group-aware adaptive assessment through modular services that integrate smoothly with existing eLearning platforms. The work pursues four main objectives:

1. *Design and implementation*: Specify and deploy an SOA-based framework (SOFIA) for real-time adaptive assessment in collaborative eLearning.
2. *Algorithms*: Formalize context-aware analytics for engagement measurement, priority-based task allocation, and adaptive feedback generation.
3. *Evaluation*: Examine effects on assessment accuracy, learner engagement, and system scalability relative to static and rule-based baselines.
4. *Implications*: Analyze pedagogical and technological implications for educators, developers, and institutions.

The following research questions guide this study:

- *RQ1*: Does SOFIA improve learning outcomes and engagement relative to static and rule-based systems?
- *RQ2*: How consistent are SOFIA's adaptive decisions with target difficulty–proficiency alignment?
- *RQ3*: How does SOFIA perform under increasing concurrent usage in terms of latency and reliability within an LMS-integrated environment?

Addressing these research questions provides a structured foundation for examining SOFIA's pedagogical and technical impact, ensuring coherence between system design, experimental evaluation, and interpretation of findings.

The remainder of this article is organized as follows. Section 2 reviews existing eLearning assessment methodologies and identifies the key gaps addressed by SOFIA. Section 3 presents the conceptual framework and design principles, emphasizing its modular architecture and integration strategies. Section 4 details the implementation workflow and adaptive algorithm. Section 5 reports the experimental evaluation conducted in the online Python programming course. Section 6 compares SOFIA's performance with traditional and rule-based adaptive systems. Section 7 discusses the findings, implications, and limitations of this research, and Sect. 8 concludes with perspectives for future work.

Literature review

The evolution of eLearning, driven by rapid advancements in technology, has created a pressing need to rethink traditional assessment methodologies. The challenge lies in addressing the dynamic and collaborative nature of modern educational environments while leveraging technological innovations to enhance learning outcomes. This review examines current assessment strategies, identifies their limitations, and situates the SOFIA within broader trends in adaptive learning and AI.

Current eLearning assessment solutions

Traditional eLearning assessments predominantly employ static frameworks, such as standardized tests, quizzes, and assignments, that are often designed for linear and individual-focused learning. These approaches, while widely used, fail to account for the immediacy and adaptability required to address the needs of contemporary learners (Pellegrino et al., 2001). They frequently result in delayed feedback, impeding timely interventions critical for enhancing the learning process (Black & Wiliam, 2009).

Moreover, these frameworks inadequately support collaborative learning, where peer interactions and group dynamics are vital to knowledge acquisition (Johnson & Johnson, 2009). Their rigidity and lack of customization often lead to disengagement and suboptimal performance, underscoring the need for more responsive assessment tools (Baidoo-Anu & Owusu Ansah, 2023).

To address these limitations, researchers have explored alternative methodologies such as formative assessments, peer feedback systems, and learning analytics. While these approaches offer valuable insights into learner behaviors, they frequently require substantial infrastructure, educator training, or are hindered by objectivity and scalability

challenges (Muslim et al., 2022). For instance, adaptive learning systems using rule-based mechanisms can tailor feedback but lack real-time responsiveness and scalability (Bodily & Verbert, 2017). Table 1 provides a comparative analysis of existing methodologies, highlighting how SOFIA addresses these gaps by integrating predictive analytics and real-time feedback mechanisms.

Advances in AI and service-oriented architectures in education

Recent advancements in AI have accelerated the development of adaptive assessment frameworks capable of delivering personalized learning experiences at scale. Machine learning models now enable real-time processing of large, heterogeneous datasets, supporting dynamic assessment and feedback tailored to both individual and group needs (Hadyaoui & Cheniti-Belcadhi, 2023). These innovations play a critical role in fostering equitable and inclusive learning environments, as emphasized by Dumont and Ready (2023).

In parallel, SOA has emerged as a transformative approach for ensuring modularity, scalability, and interoperability in educational systems. The microservices paradigm central to SOA allows for the seamless integration of new tools and continuous updates without disrupting existing infrastructure. Prior studies highlight the scalability and adaptability of SOA in supporting diverse learner needs across contexts (Dragoni et al., 2017; Papazoglou et al., 2018).

SOFIA builds on these technological and pedagogical advancements through the combination of AI-driven analytics and SOA principles, resulting in a robust and scalable adaptive assessment framework. Its architecture supports real-time data collection, priority-based task optimization, and dynamic feedback generation, ensuring alignment with evolving learner profiles and interaction patterns. Unlike static or rule-based adaptive systems, SOFIA integrates predictive analytics to continuously adjust assessments in response to learner progress and group dynamics, enabling timely and targeted interventions.

Although many adaptive learning frameworks integrate AI, they often lack real-time responsiveness and scalability for collaborative learning contexts (Ezzaim et al., 2023). Traditional adaptive systems tend to prioritize individual learning trajectories, offering limited support for group-based assessment and complex interaction patterns. SOFIA addresses these limitations through asynchronous processing and a priority-based task optimization algorithm that jointly accounts for individual and group needs.

SOFIA's contributions reflect broader educational trends that emphasize personalization, learner engagement, and equity. The integration of predictive analytics and real-time feedback loops enhances learning outcomes while addressing systemic challenges such as scalability, inclusivity, and the incorporation of collaborative pedagogical practices.

Framework design

The Service-Oriented Framework for Intelligent Assessment (SOFIA) addresses challenges in collaborative eLearning by combining SOA with AI-driven analytics. The framework delivers real-time, group-aware adaptive assessment for individuals and teams. As shown in Fig. 1, SOFIA adopts a modular microservices design that

Table 1 Comparative table: existing elearning assessment methods versus SOFIA

Feature	Traditional assessments	Formative assessments	Peer assessment	Adaptive systems	SOFIA
Personalization	Minimal, static, one-size-fits-all	Adaptive to some extent but lacks scalability	Feedback depends on peer input, limited adaptability	Provides tailored feedback using AI but often costly	Highly personalized, using predictive analytics and real-time data
Scalability	Limited to manual processes and small groups	Scalable with educator effort but resource-intensive	Scalable but struggles with maintaining objectivity	Partially scalable; requires significant infrastructure	Fully scalable through SOA and microservice architecture
Real-time feedback	Absent, delayed until post-assessment	Provides real-time insights but requires heavy setup	Feedback timing depends on peers	Limited real-time capabilities in many implementations	Dynamic and instantaneous feedback leveraging AI
Integration with LMS	Basic integration, often rigid	Can integrate but requires customization	Limited integration; manual processes predominate	Partial integration, often with proprietary systems	Seamless integration via xAPI, SCORM, and LTI standards
Ease of maintenance	Requires frequent manual intervention	Moderate, dependent on educator involvement	High effort for consistent quality control	Requires regular updates and specialized tools	Low-maintenance modular design enabled by SOA
Cost Efficiency	Low cost but limited functionality	High resource and training costs	Low cost but depends on student initiative	High development and operational costs	Cost-effective due to modular, reusable components
Group dynamics support	Absent	Limited	Strong focus on collaboration but lacks automation	Limited group support, focused on individuals	Supports individual and group assessments with AI
Engagement and interaction	Limited, passive assessments	Moderate, depends on implementation	High engagement through peer interaction	Moderate engagement, focused on individual adaptation	High engagement through adaptive, collaborative scenarios

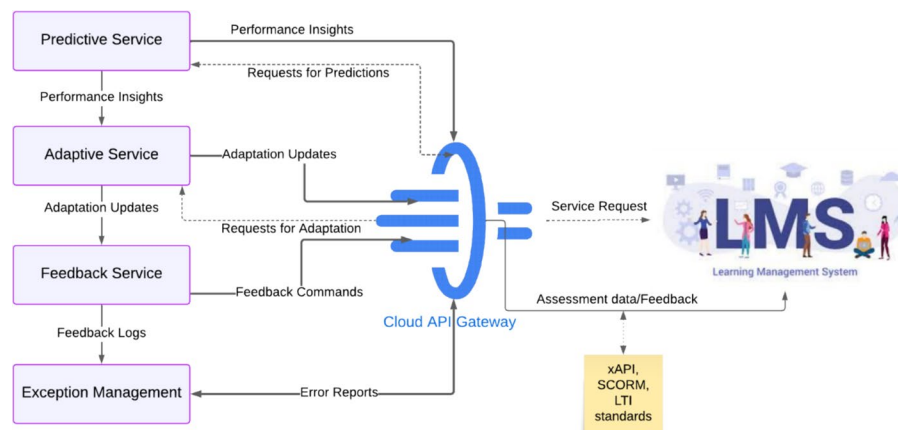


Fig. 1 SOFIA system architecture (SOA-based adaptive assessment)

integrates with the existing LMS through well-defined APIs and secure event streams. Each service operates independently and communicates over a lightweight message layer, which supports deployment flexibility, resilience to partial failures, and evolution of components without disrupting the LMS. This architectural choice enables consistent interoperability across heterogeneous institutional environments while maintaining clear separation of concerns between engagement analytics, priority-based task allocation, feedback generation, and monitoring/observability.

Key design principles of the SOFIA framework

The SOFIA framework integrates AI and machine learning techniques—including predictive analytics and natural language processing (NLP)—to process learner interaction data and generate actionable insights about engagement, comprehension, and skill development. These capabilities aim to support timely, adaptive interventions in collaborative learning environments, addressing persistent challenges related to personalization, feedback latency, and scalability.

The architectural design of SOFIA is grounded in SOA principles, which enhance the system's flexibility, interoperability, and operational resilience. The following five principles guide the framework's implementation:

- *Modularity* Each core service—Predictive, Adaptive, Feedback, and Exception Management—operates independently and can be updated or scaled without disrupting the overall system. This ensures continuous operation during maintenance or iterative upgrades.
- *Loose coupling* Services communicate through well-defined APIs, minimizing dependencies between components. This design reduces system fragility and simplifies the integration of new services or algorithms.
- *Interoperability* By adhering to standard communication protocols (e.g., RESTful APIs, xAPI, SCORM, LTI), SOFIA integrates seamlessly with various LMS and external educational technologies.

- *Reusability* Modular components can be redeployed across different learning contexts and disciplines, increasing implementation efficiency and reducing development overhead.
- *Discoverability* A service registry in the Cloud API Gateway enables automatic service discovery, dynamic configuration, and runtime scaling. This supports real-time load management and aligns with the framework's goal of institutional-level deployment.

Together, these principles provide the technical foundation for SOFIA's real-time adaptive capabilities, ensuring the framework can support RQ1 (learning outcomes and engagement), RQ2 (adaptive decision accuracy), and RQ3 (system scalability and reliability).

Key architectural components

The SOFIA framework integrates a set of modular, service-oriented components, each contributing to the framework's real-time adaptivity, scalability, and interoperability. These services work collaboratively through the Cloud API Gateway to support adaptive assessment in collaborative eLearning environments:

- *Predictive service* Collects and processes learner interaction data to generate actionable insights through predictive analytics. This service estimates learner proficiency, engagement trends, and potential misconceptions, forming the basis for adaptive task allocation (supports RQ2).
- *Adaptive service* Dynamically assigns tasks and adjusts assessment parameters in response to real-time predictive insights. By aligning task difficulty with learner proficiency and group context, it personalizes learning experiences and supports adaptive decision-making (supports RQ1 and RQ2).
- *Feedback service* Delivers targeted, context-sensitive feedback to learners and groups during the learning process. This service integrates output from the predictive and adaptive components to enhance engagement and facilitate formative learning cycles (supports RQ1).
- *Exception management service* Monitors service interactions and operational anomalies, ensuring system stability and fault tolerance. It enables graceful handling of disruptions without interrupting learner activities (supports RQ3).
- *Integration service* Facilitates secure and seamless data exchange between SOFIA and external platforms, such as Learning Management Systems (e.g., Moodle). This service leverages interoperability standards including xAPI, SCORM, and LTI to ensure compatibility with a wide range of educational ecosystems (supports RQ3).

Microservices interaction via the cloud API gateway

The SOFIA framework employs a microservices architecture in which each service operates independently but communicates through a central Cloud API Gateway (Fig. 2). This gateway acts as the orchestration layer, managing data transmission, routing, and service discovery between internal microservices and external systems such as LMS

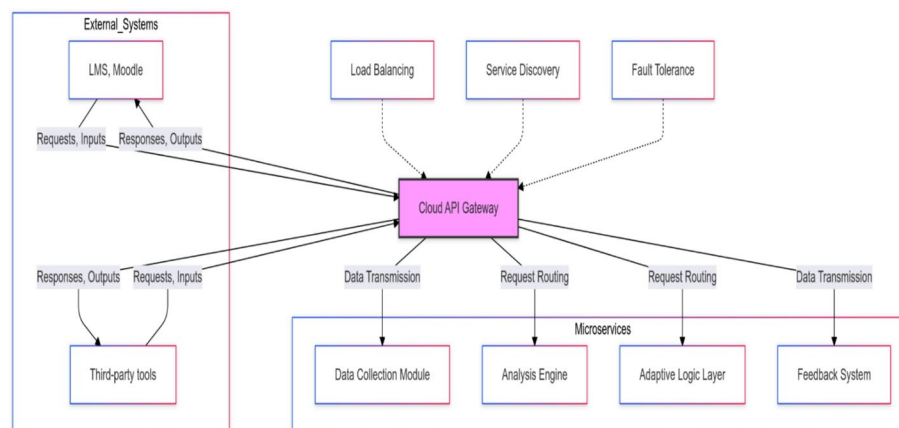


Fig. 2 Microservices integration via the cloud API gateway

and third-party tools. This architecture ensures secure, scalable, and low-latency communication—critical for real-time adaptive assessment at both the individual and group levels.

The Cloud API Gateway enables the following key interactions:

- *Request handling* The gateway receives and authenticates incoming requests from LMSs and external tools (e.g., data retrieval, assessment updates). It standardizes requests before routing them to the appropriate microservice, ensuring secure and consistent communication (supports RQ1 and RQ3).
- *Service discovery* A dynamic service registry allows the gateway to detect and access available microservices—including the Data Collection Module, Analysis Engine, Adaptive Logic Layer, and Feedback System. This mechanism enables horizontal scaling and component updates without system downtime (supports RQ3).
- *Routing* Incoming requests are routed to the relevant microservice using predefined APIs. For example, performance data requests are directed to the Data Collection Module, while adaptive task allocation requests are routed to the Adaptive Logic Layer (supports RQ1 and RQ2).
- *Data flow* Bidirectional communication ensures that outputs from one microservice (e.g., learner analytics from the Analysis Engine) are efficiently passed to others (e.g., the Adaptive Logic Layer for personalized task allocation), supporting coordinated, real-time adaptation (supports RQ1 and RQ2).
- *Load balancing and fault tolerance* The gateway distributes traffic evenly across services to maintain performance under variable loads. In case of failure, requests are rerouted to redundant service instances, ensuring system resilience and uninterrupted learner experiences (supports RQ3).

Service-oriented architecture (SOA) principles in SOFIA

The implementation of SOFIA is grounded in core SOA principles, which provide the foundation for its flexibility, interoperability, and scalability in dynamic educational settings. These principles—service discovery, scalability, and integration with LMS

platforms—enable SOFIA to operate efficiently, respond to changes in real time, and support adaptive assessment for individuals and collaborative groups. Their application directly supports the framework’s capacity to handle large-scale deployment (RQ3), maintain adaptive responsiveness (RQ2), and ensure seamless learning ecosystem integration (RQ1 and RQ3).

Table 2 summarizes how each SOA principle is implemented within SOFIA and provides illustrative use cases from real-world learning scenarios.

Implementation

The implementation of SOFIA translates its service-oriented architectural design into a fully operational system within an institutional learning environment. This implementation enables real-time, group-aware adaptive assessment and is structured to support RQ1 (learning outcomes and engagement), RQ2 (adaptive decision accuracy), and RQ3 (scalability and reliability). The following subsections describe the system setup, operational workflow, and integration processes that ensure seamless deployment.

System setup

The setup of SOFIA follows a structured configuration process designed to ensure secure and efficient integration with existing LMSs. Its modular architecture supports flexible deployment across different institutional infrastructures.

- *Infrastructure preparation* Implementation begins with establishing the necessary digital infrastructure, including provisioning servers, configuring databases, and ensuring network capacity to support high-throughput data exchange and low-latency communication.
- *Software deployment* Once the infrastructure is in place, the Data Collection Module, Analysis Engine, Adaptive Logic Layer, and Feedback System are deployed as independent microservices. These components are interconnected through the Cloud API Gateway, enabling secure and efficient communication between services and external systems.
- *Integration configuration* SOFIA is linked to the LMS via API endpoints using standardized interoperability protocols (xAPI, SCORM, and LTI). This step ensures secure data exchange, standardized message formats, and real-time synchronization of assessment results and feedback within the LMS environment.

Ethical compliance

All participants provided informed consent for the use of their anonymized learning data. Data collection and analysis were conducted in accordance with the ethical standards of the University of Sousse and complied with national and international data protection regulations, including GDPR principles. The study design and implementation respected institutional review requirements for educational research.

Table 2 Key SOA principles applied

SOA principle	Description	Implementation	Example use case
Service discovery	Dynamic identification and invocation of services during runtime, enabling flexible reconfiguration	Services (e.g., <i>Data Collection Module</i> , <i>Analysis Engine</i>) are registered with metadata (functionality, endpoints) The Cloud API Gateway queries the registry to route requests	When a new <i>Data Analytics Service</i> is introduced during an ongoing assessment, the service registry makes it discoverable and operational without downtime
Scalability	The architecture can handle increasing resource demands or user loads without performance degradation	Horizontal scaling through dynamic service instantiation Load balancing distributes traffic Stateless services (e.g., <i>Analysis Engine</i>) ensure elasticity	During peak demand (e.g., final exams), SOFIA launches additional instances of critical services like the <i>Analysis Engine</i> to maintain real-time feedback without latency
Integration with LMS	Ensures interoperability with external platforms, enabling data flow between SOFIA and existing systems	Cloud API Gateway manages secure bidirectional communication with Moodle via xAPI, SCORM, and LTI Results and feedback are pushed to the LMS gradebook	A learner submits a programming assignment on Moodle. SOFIA retrieves it via xAPI, analyzes performance through the <i>Adaptive Logic Layer</i> , and returns personalized feedback directly to the Moodle interface

Microservices interactions via the cloud API gateway

The Cloud API Gateway plays a central role in the SOFIA framework by orchestrating secure, low-latency communication between internal microservices and external systems such as LMS. Its design enables real-time adaptive assessment at scale, supporting both RQ2 (adaptive decision accuracy) and RQ3 (system scalability and reliability).

The gateway ensures reliable routing, dynamic service discovery, performance optimization, and fault tolerance, even under high user loads.

Key features and functionalities

- *Request handling and security* All external requests are authenticated through secure protocols (e.g., OAuth 2.0, JWT) to ensure controlled access. Rate limiting regulates traffic during peak periods, preventing gateway saturation and maintaining consistent response times.
- *Dynamic service discovery* The gateway uses service registries (e.g., Eureka, Consul) to locate and invoke microservices such as the Analysis Engine and Feedback System at runtime. This mechanism allows services to be updated or replaced without downtime, improving operational flexibility and scalability.
- *Routing and traffic management* Advanced routing rules direct incoming traffic to the appropriate service (e.g., data processing requests to the Analysis Engine, feedback generation to the Feedback System). Routing strategies (path-based or host-based) minimize latency and optimize throughput.
- *Load balancing* Algorithms such as round-robin and least-connections distribute requests evenly across service instances. This ensures optimal resource utilization and mitigates performance bottlenecks during concurrent access.
- *Caching and data optimization* Frequently accessed resources (e.g., assessment results) are cached using in-memory data stores like Redis or Memcached, while compression (e.g., GZIP) reduces bandwidth consumption and speeds up responses.
- *Monitoring and fault tolerance* Real-time monitoring (e.g., Prometheus, Grafana) tracks system health and performance. Circuit breakers (e.g., Hystrix) detect failures and trigger fallback mechanisms to maintain system stability.

Performance optimization strategies under high user loads

- *Horizontal scalability* Each microservice can scale independently. For example, the Analysis Engine can add instances to process large data volumes, while the Adaptive Logic Layer scales to maintain response time for personalized feedback.
- *Asynchronous communication* Message queues such as Apache Kafka or RabbitMQ decouple services, ensuring that high-volume requests do not block critical operations. This supports non-blocking processing and smooth throughput.
- *API gateway enhancements* Multi-threading and connection pooling increase gateway throughput during concurrent user sessions. Intelligent traffic shaping dynamically allocates resources based on user activity patterns.

- *Distributed architecture* Container orchestration platforms (e.g., Kubernetes) manage deployment across distributed nodes. This architecture minimizes downtime during maintenance or scaling events and guarantees high availability.

Operational flow

SOFIA operates through a real-time adaptive workflow that continuously responds to learner interactions, supporting RQ1 (learning engagement and outcomes) and RQ2 (adaptive decision accuracy). As shown in Fig. 3, this operational process ensures that assessment and feedback remain dynamically aligned with learner progress.

- *Step 1: Data collection.* Learner interactions with the LMS trigger real-time data acquisition through the Data Collection Module. Metrics such as engagement level, interaction duration, and task performance are captured, forming the foundation for analysis.
- *Step 2: Data processing and analysis.* The Analysis Engine processes the collected data, applying algorithms to infer learning behaviors and performance patterns. This analysis identifies strengths, misconceptions, and knowledge gaps at both the individual and group levels.
- *Step 3: Adaptation and feedback planning.* Insights generated by the Analysis Engine are transmitted to the Adaptive Logic Layer, which dynamically adjusts assessment parameters and learning content. This ensures that each learner receives appropriately challenging tasks that match their evolving proficiency.
- *Step 4: Continuous feedback delivery.* The Feedback System communicates targeted guidance and personalized recommendations back to the learner through the LMS. This feedback loop supports iterative learning, enabling timely correction and reinforcement of concepts.



Fig. 3 SOFIA adaptive assessment cycle

Together, these steps form a closed adaptive loop that maintains alignment between learner activity, assessment content, and feedback. This design allows SOFIA to respond dynamically to changing learner needs, ensuring personalization at scale.

Dashboard and analytics

The integrated SOFIA Dashboard acts as a central decision-support interface for educators and administrators, enabling real-time personalization at scale. It aggregates and visualizes outputs generated in Steps 2 to 4 of the adaptive algorithm—namely, *engagement and accuracy indicators*, *priority-based task assignments*, and *adaptive feedback signals*. This integration allows for continuous monitoring, early detection of learning challenges, and data-informed pedagogical decision-making.

As shown in Fig. 4, the dashboard presents key performance indicators through interactive and intuitive visualizations. These include:

- Engagement trends over time,
- Task completion rates, and
- Accuracy distributions across learners and groups.

These visual insights allow instructors to rapidly identify learning patterns, detect anomalies, and implement targeted interventions that support both individual progress and collaborative group dynamics.

Beyond descriptive analytics, the analytical engine incorporates statistical inference and machine learning models—including decision trees, neural networks, and support vector machines—to process high-volume, complex learner data. These models:

- Identify early warning signals such as disengagement or performance decline,
- Highlight emerging knowledge gaps and skill bottlenecks, and
- Recommend targeted pedagogical responses aligned with adaptive task allocation.

This predictive capability enables personalized interventions at scale, allowing instructors to act before performance declines become critical. The integration of real-time visualization with predictive analytics strengthens the human-in-the-loop adaptive cycle, ensuring transparency, scalability, and actionable insights.



Fig. 4 SOFIA analytics dashboard: tracking learner engagement and completion metrics over time

The real-time adaptive feedback generation algorithm in SOFIA

Overview and operational workflow

The SOFIA adaptive assessment engine is designed to personalize task allocation and feedback generation in real time, based on learner interaction and performance patterns. Its architecture integrates five interconnected components: (1) data collection and preprocessing, (2) computation of engagement and accuracy indicators, (3) priority-based task optimization, (4) adaptive feedback generation, and (5) feedback delivery and session logging. This modular structure ensures interpretability, scalability, and adaptability to various instructional contexts. Figure 5 illustrates the operational workflow, while Algorithm 1 formalizes the procedural logic.

Step 1—Data collection and preprocessing.

Learner interaction data are ingested in real time from the LMS. The system records task completion time, error counts, interaction frequency, and time-on-task. Records undergo validation and normalization to ensure data consistency.

Step 2—Engagement and accuracy analytics.

The analysis engine computes task-level engagement as the ratio of interaction frequency to time-on-task, and task-level accuracy as the proportion of correct responses. Learner-level engagement and accuracy are aggregated over all assigned tasks. Thresholds for low engagement ($\tau_{\text{eng}} = 0.10$) and low accuracy ($\tau_{\text{acc}} = 0.80$) trigger corresponding flags, which are used in subsequent decision steps to adapt task allocation and feedback.

Step 3—Priority-based task optimization.

A priority score $\pi(\ell, t) = |P_\ell - D_t|$ is computed for each learner ℓ and task t . A min-heap priority queue ensures that tasks best aligned with a learner's proficiency are selected first. Deterministic tie-breaking is applied using task ID and minimal expected time-on-task.

Step 4—Adaptive feedback generation.

Feedback and learning recommendations are tailored to the learner's engagement and performance profile. Low-accuracy learners receive error-reduction guidance, low-engagement learners are assigned motivational prompts and strategically selected high-yield tasks, and learners without engagement or accuracy flags are encouraged to attempt stretch tasks that moderately exceed their proficiency levels.

Step 5—Feedback delivery and logging.

The generated recommendations are delivered to the learner's dashboard and integrated into the LMS gradebook. All computed indicators, task assignments, and feedback outputs are logged to support monitoring, instructor dashboards, and future refinement of adaptive decision rules.

Parameterization

To ensure reproducibility, all algorithmic variables, thresholds, and outputs are explicitly defined in Table 3. Parameters include learner proficiency (P_ℓ), task difficulty (D_t), engagement (E_ℓ), accuracy (Acc_ℓ), and task priority scores ($\pi(\ell, t)$).

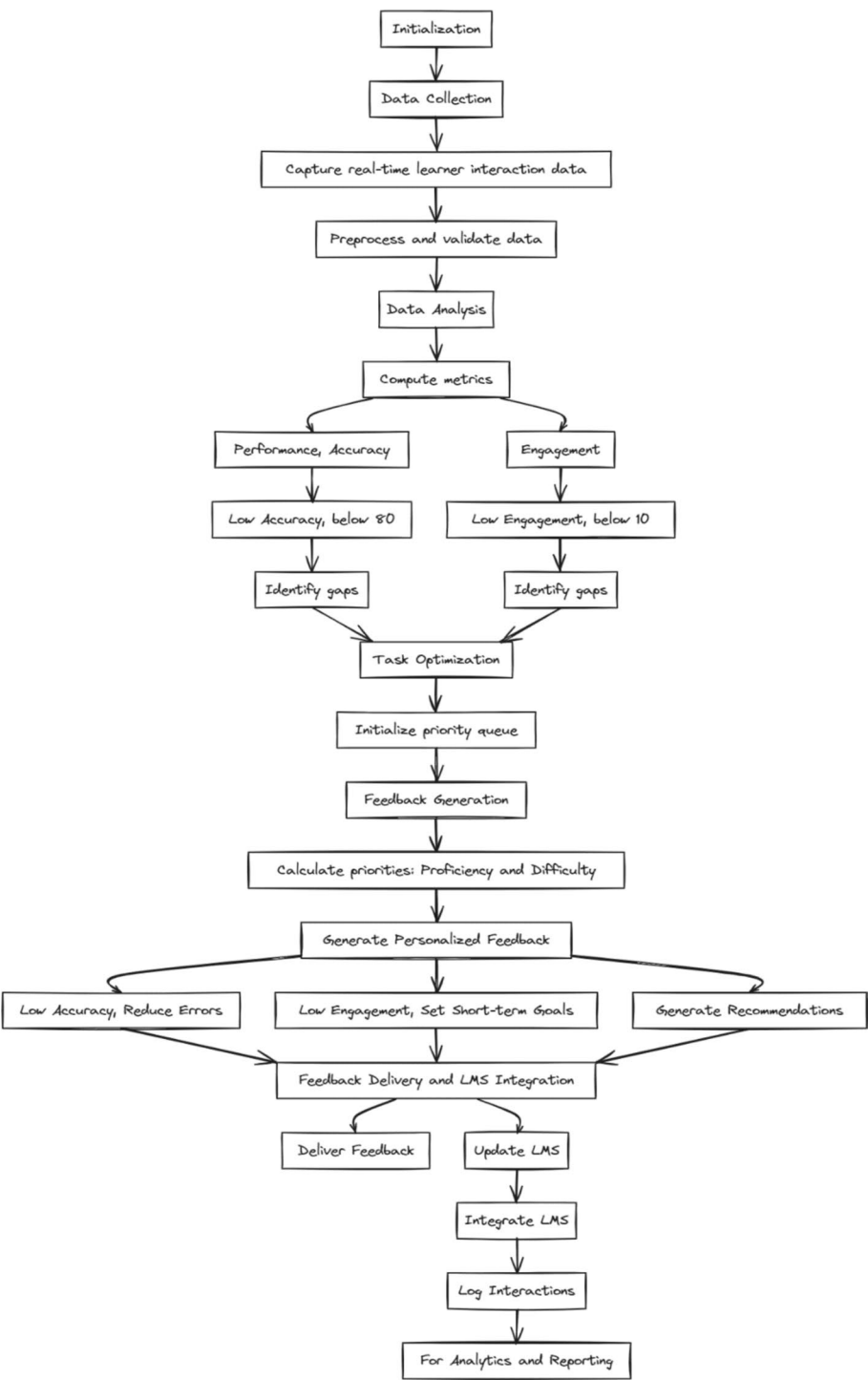


Fig. 5 Real-time adaptive feedback generation workflow in SOFIA

Threshold values for engagement and accuracy were determined through empirical tuning and can be adapted to different instructional settings. Logged outputs include

Table 3 Summary of variables, parameters, and notations used in the SOFIA adaptive assessment algorithm

Symbol/variable	Description	Type/range	Default/example value
Input parameters	(used for data collection and preprocessing)	–	–
$\mathcal{L}$	Set of learners in the session	Set	–
$\mathcal{T}$	Set of tasks available for allocation	Set	–
P_ℓ	Proficiency level of learner ℓ	Real number [0, 1]	Simulated or estimated
D_t	Difficulty level of task t	Real number [0, 1]	Uniform distribution
$IF_{\ell,t}$	Interaction frequency for learner ℓ on task t derived from platform interaction logs	Integer (≥ 0)	Derived from LMS logs
$ToT_{\ell,t}$	Time on task for learner ℓ on task t	Real number (> 0)	Measured in seconds or minutes
Engagement and accuracy metrics	(computed during analytics & flagging)	–	–
$E_{\ell,t}$	Engagement per task $IF_{\ell,t}/ToT_{\ell,t}$	Real number (> 0)	Computed
E_ℓ	Session-level engagement for learner ℓ : $\frac{\sum_t IF_{\ell,t}}{\sum_t ToT_{\ell,t}}$	Real number (> 0)	Aggregated
Acc_ℓ	Session-level accuracy for learner ℓ computed as correct / attempted responses	Real number [0, 1]	Computed
Thresholds	(used to trigger flags)	–	–
τ_{acc}	Low accuracy threshold	Real number [0, 1]	0.80
τ_{eng}	Low engagement threshold	Real number (> 0)	0.10
Priority and Flags	(used for task assignment and feedback)	–	–
$\pi(\ell, t)$	Task priority score $ P_1 - D_t $	Real number (≥ 0)	Computed
Q	Priority queue (min-heap) containing (π, tie_key, ℓ, t)	Data structure	—
$flagLowAcc_\ell$	Indicator for low accuracy	Boolean	$Acc_\ell < \tau_{acc}$
$flagLowEng_\ell$	Indicator for low engagement	Boolean	$E_\ell < \tau_{eng}$
tie_key	Tie-breaking key	ID/time	
Outputs and logged variables	(used for feedback, LMS update & analytics)	–	–
PF_ℓ	Personalized feedback message for learner ℓ	Text	Generated by the algorithm
URL_ℓ	Updated learning recommendations for learner ℓ	List of tasks	Generated by the algorithm
$A[\ell]$	Final task assignment for learner ℓ	Task ID	Computed from Q
Logs	Logged variables: $\pi(\ell, t), E_\ell, Acc_\ell, flagLowAcc_\ell, flagLowEng_\ell, A[\ell], A[\ell], URL_\ell$	–	–

all computed scores, flag values, and feedback content, enabling both real-time and retrospective analysis.

real-time adaptive feedback, task optimization, and delivery (SOFIA) algorithm

Algorithm 1 Real-time adaptive feedback, task optimization, and delivery (SOFIA)

Inputs:

- LMS event stream;
- Learner profiles P_ℓ ;
- Task set D_t ;
- Thresholds $\tau_{acc} = 0.80$, $\tau_{eng} = 0.10$

Outputs:

- PF_ℓ : Personalized feedback,
- URL_ℓ : Updated learning recommendations,
- $A[\ell]$: Optimized task assignments

Step 1. Data Collection & Preprocessing

1. Ingest LMS events for each learner ℓ and task t
2. Validate records and compute:
 - $IF_{\ell,t}$: interaction frequency
 - $ToT_{\ell,t}$: time on task
 - task-level accuracy $Acc_{\ell,t}$
3. Aggregate interaction data across tasks to compute session-level engagement E_ℓ and accuracy Acc_ℓ for each learner.

Step 2. Analytics & Flagging

1. Compute engagement for each learner:

$$E_\ell = \frac{\sum_t IF_{\ell,t}}{\sum_t ToT_{\ell,t}}$$

2. Compute session-level accuracy Acc_ℓ
3. If $Acc_\ell < \tau_{acc}$, set $flagLowAcc_\ell = True$
4. If $E_\ell < \tau_{eng}$, set $flagLowEng_\ell = True$

Step 3. Priority-Based Task Optimization

1. For each learner ℓ :
 - a. Compute $\pi(\ell, t) = |P_\ell - D_t|$
 - b. Push (π, tie_key, ℓ, t) into priority queue Q
2. While Q is not empty:
 - a. Pop the element with minimal priority π .
 - b. If learner ℓ has no assigned task and t is available, set $A[\ell] = t$.
3. Tie-breaking rule: If two tasks have equal priority, select the task with
 - a. lowest task ID, or if still tied,
 - b. the shortest expected time-on-task.

Step 4. Adaptive Feedback Generation

1. Initialize PF_ℓ and ULR_ℓ for each learner.
2. If $\text{flagLowAcc}_\ell = \text{True}$:
 - Add feedback: “Focus on error reduction.”
 - Recommend targeted tasks addressing weak areas.
3. If $\text{flagLowEng}_\ell = \text{True}$:
 - Add feedback: “Increase participation; aim for short-term goals.”
 - Recommend short, high-yield tasks to boost activity.
4. If both flags are raised:
 - Concatenate both messages and prioritize motivational tasks first.
5. If no flags are raised:
 - add feedback: “Maintain current pace and attempt one stretch task.”
6. Append $A[\ell]$ and a set of alternate tasks to ULR_ℓ .

Step 5. Feedback Delivery and LMS Update

1. Deliver PF_ℓ and ULR_ℓ to each learner dashboard.
2. Update LMS gradebook and task assignment records with $A[\ell]$.

Step 6. Logging and Analytics

1. Log for each learner:

$$\pi(\ell, t), E_\ell, \text{Acc}_\ell, \text{flagLowAcc}_\ell, \text{flagLowEng}_\ell, A[\ell], PF_\ell, ULR_\ell.$$
2. Store logs for performance monitoring, analytics dashboards, and system refinement.

This pseudocode formalizes the adaptive process, providing traceability between learner interaction data, algorithmic decisions, and personalized outputs.

Worked example

To illustrate the priority-based allocation mechanism, consider a learner with proficiency $P_\ell = 0.7$. Two candidate tasks have difficulty values $D_{t1} = 0.6$ and $D_{t2} = 0.2$.

$$\pi(\ell, t_1) = |0.7 - 0.6| = 0.1$$

Compared to another task with difficulty 0.2:

$$\pi(\ell, t_2) = |0.7 - 0.2| = 0.5$$

Since $0.1 < 0.5$, Task t_1 is prioritized because it offers the closest match between task difficulty and learner proficiency, ensuring appropriate challenge and sustained engagement.

Engagement metric justification

Engagement is operationalized as the ratio of interaction frequency to total time-on-task for each learner:

$$E_{\ell} = \frac{\sum_t IF_{\ell,t}}{\sum_t ToT_{\ell,t}}$$

where $IF_{\ell,t}$ represents the interaction frequency of learner ℓ on task t , and $ToT_{\ell,t}$ corresponds to the time spent on the same task.

This formulation reflects interaction intensity relative to time and provides a normalized measure that can be compared across learners and sessions. Similar engagement metrics have been widely adopted in learning analytics research to identify behavioral patterns and detect low participation (e.g., Shute, 2011; Baker & Inventado, 2014).

Sensitivity illustration

- Learner A: 20 interactions / 10 minutes $\rightarrow E = 2.0$
- Learner B: 20 interactions / 40 minutes $\rightarrow E = 0.5$

Although both learners exhibit the same number of interactions, the metric differentiates their engagement intensity. Learner A is flagged as more active due to higher interaction frequency within a shorter time span.

Limitations

This metric captures behavioral engagement intensity but does not account for interaction quality or cognitive engagement. High-frequency superficial interactions may artificially inflate engagement scores, whereas fewer but meaningful contributions could be undervalued.

To address this limitation, future work will incorporate multimodal indicators, including semantic analysis of learner contributions and affective signals, to enhance the robustness and depth of engagement modeling.

Task priority

Task prioritization seeks to align learner proficiency with task difficulty, thereby maintaining an optimal cognitive load. The current formulation is expressed as:

$$\pi(\ell, t) = |P_{\ell} - D_t|$$

where P_{ℓ} denotes the proficiency of learner ℓ and D_t represents the difficulty of task t .

This function is transparent, computationally efficient, and easily interpretable by educators and system developers. Tasks with lower priority values are assigned first, ensuring learners receive activities that are appropriately challenging without being either trivial or overwhelming.

For example, a learner with $P_{\ell} = 0.7$ will prioritize tasks with D_t values closest to 0.7, reflecting the best proficiency–difficulty match.

While effective as a baseline, this formulation can be extended with motivational weighting factors or collaborative dynamics to account for additional variables such as learner persistence, affective states, or group composition effects. These extensions would allow the system to support more complex and adaptive learning contexts.

Simulation

The dataset used for evaluating SOFIA integrates authentic learner interaction traces with synthetic extensions generated under controlled conditions. Real interaction data, collected during the experimental deployment, provided the empirical foundation for modeling engagement and performance patterns.

To enable systematic scalability testing, the dataset was algorithmically extended to 500 learners and 120 tasks. Learner proficiency levels were sampled from a normal distribution, task difficulties from a uniform distribution, interaction frequencies from a Poisson distribution, and completion times from a log-normal distribution. These choices reflect empirically observed patterns in learner interaction data, including positively skewed completion times and count-based interaction frequencies. Error rates were computed based on proficiency–difficulty gaps, ensuring the preservation of realistic behavioral dynamics.

All data generation and algorithm evaluation scripts were implemented in Python 3.10 with fixed random seeds to ensure full reproducibility. This simulation framework allows systematic evaluation of SOFIA's task allocation and adaptive feedback mechanisms under varying learner and task conditions.

Methodology

The methodology adopted in this study was carefully structured to ensure experimental rigor, fairness, and reproducibility. It details the participant allocation, evaluation metrics, control measures, ethical considerations, and experimental setup for assessing SOFIA's pedagogical and technical performance.

Participant selection and group allocation

A total of 250 undergraduate students enrolled in a Python programming course participated in the study. The selection process ensured a balanced representation of skill levels, academic backgrounds, and socioeconomic demographics to enhance inclusivity and generalizability. Eligibility criteria included regular attendance, completion of prerequisite modules, and consent to participate in the experiment.

Participants were randomly allocated in approximately equal numbers between the experimental and control groups, ensuring balanced group sizes for valid comparison. The experimental group used SOFIA for adaptive assessments, while the control group relied on traditional assessment methods. Stratified randomization was applied to maintain balance between groups in terms of initial proficiency levels, engagement patterns, and demographic characteristics. The final sample size was determined through a power analysis, which ensured sufficient statistical power to detect meaningful differences in learning outcome.

Metrics for evaluation

The evaluation framework combined quantitative and qualitative metrics to capture both cognitive and behavioral dimensions of learning. This multidimensional approach ensured a comprehensive assessment of learner outcomes, engagement patterns, and skill transfer in relation to SOFIA's adaptive interventions.

- *Critical thinking scores*: Pre- and post-assessment tasks were administered to evaluate students' ability to transfer and apply acquired knowledge in unfamiliar problem contexts. A structured rubric assessed dimensions of problem-solving, reasoning, and creativity, ensuring consistency and comparability across groups.
- *Engagement levels*: Clickstream data from the LMS were analyzed to extract task completion rates, interaction frequency, and time-on-task. These indicators distinguished active engagement from passive participation, aligning with the engagement analytics operationalized in the adaptive algorithm (Sect. 4.4).
- *Learning retention rates*: Follow-up quizzes conducted two weeks after the final assessment measured long-term knowledge retention and independent application of skills. This metric provided insight into the sustained impact of adaptive versus traditional assessment approaches.
- *Task performance*: Practical programming tasks were evaluated using accuracy rates, debugging efficiency, and task completion time. These indicators reflected students' applied problem-solving abilities and operationalized the performance dimensions modeled in SOFIA's adaptive feedback algorithm.

Control variables and bias mitigation

Several control measures were implemented to strengthen the internal validity, reliability, and interpretive integrity of the experimental findings. These measures aimed to reduce confounding factors and ensure that observed differences in learning outcomes could be attributed to the SOFIA intervention rather than uncontrolled external influences.

- *Lecturer involvement*: To minimize potential bias stemming from the lecturer's dual role as instructor and researcher, all assessment data were anonymized prior to analysis. An independent evaluator reviewed engagement logs and performance outcomes to ensure impartial interpretation. In addition, a heuristic usability evaluation conducted by three domain experts provided qualitative feedback on the dashboard's clarity, interpretability, and perceived neutrality.
- *Environmental consistency*: Both the experimental and control groups received identical teaching materials, schedules, and access to learning resources. This controlled for environmental and instructional variables unrelated to the intervention, ensuring that differences in outcomes reflected the impact of SOFIA rather than pedagogical inconsistencies.
- *Baseline comparisons*: Baseline quizzes were administered prior to the intervention to assess initial performance levels for both groups. These results served as

reference points to measure learning gains and mitigate the influence of pre-existing differences in skills or knowledge.

- *Mitigating external influences:* Regular monitoring was conducted to detect and minimize potential external influences such as peer effects, technological disruptions, or variations in instructional delivery. This step enhanced the attributional validity of the findings, allowing more confident interpretation of SOFIA's effects on learner engagement and performance.

Ethical considerations

The study adhered to recognized ethical standards to safeguard participant rights, ensure data integrity, and promote responsible research practices. Ethical measures were applied throughout all stages of the research process—from recruitment to data analysis and reporting.

- *Informed consent:* Participants were clearly briefed on the objectives, methodology, and scope of the study prior to providing informed consent. Participation was voluntary, and students were informed of their right to withdraw at any time without penalty or academic consequence.
- *Data privacy and anonymization:* All personally identifiable information was anonymized prior to analysis. Data collection, storage, and processing complied with institutional and international data protection principles, ensuring confidentiality, secure access, and minimal risk of disclosure.
- *Ethical approval and compliance:* The study design and procedures were reviewed and approved in accordance with Institutional Review Board (IRB) guidelines and broader scholarly ethics protocols. This process ensured the research conformed to principles of beneficence, non-maleficence, autonomy, and justice.

Experiment setup

A controlled experimental study was designed to evaluate the pedagogical and technical performance of the SOFIA algorithm using a hybrid dataset combining authentic and simulated learner interaction traces. This dual structure supports both the evaluation of adaptive assessment impact on learning outcomes and the benchmarking of system scalability under controlled conditions.

- *Data source*

The authentic dataset consisted of interaction traces from 250 undergraduate students enrolled in a Python programming course. The recorded variables included engagement levels, task performance, and behavioral interaction metrics during authentic learning sessions.

To evaluate SOFIA's scalability, this dataset was algorithmically extended to simulate a cohort of 500 learners, preserving the statistical distributions and behavioral patterns observed in the original dataset. This augmentation enabled systematic and reproducible testing of system performance under increasing user loads.

- *Baseline methods*

Table 4 Performance comparison of SOFIA algorithm with baseline methods

Metric	SOFIA algorithm	Static assessments	Rule-based adaptive systems
Response time (s)	0.12	0.05	0.08
Accuracy (%)	92	65	78
Engagement (%)	40% improvement	Baseline	15% improvement
Scalability	Handles 500+ users	Limited to 100 users	Handles up to 200 users

SOFIA's performance was compared with two baseline assessment approaches:

1. *Static assessment* Tasks assigned using fixed, predefined rules without adaptive mechanisms.
2. *Rule-based adaptive assessment* Tasks assigned using heuristic rules with basic adaptation but without real-time optimization or predictive modeling.

- Evaluation metrics

The evaluation relied on quantitative indicators aligned with the study's research questions (RQ1–RQ3):

1. *Response time* Average time required to generate personalized feedback and updated task recommendations.
2. *Accuracy* Proportion of correctly identified learning gaps and task–learner matches relative to ground truth labels.
3. *Engagement improvement* Relative increase in interaction frequency and time-on-task compared to baseline engagement levels.
4. *Scalability* System response and reliability under simulated loads (50, 250, and 500 learners).

- Experimental rationale

This experimental design ensured that real student data informed the pedagogical impact assessment, while simulated extensions allowed controlled stress testing of algorithmic components. The combination provided a rigorous, transparent, and reproducible foundation for evaluating both educational effectiveness and technical performance at scale.

Results and findings

Algorithm performance comparison

Table 4 summarizes the performance of the SOFIA algorithm compared with baseline assessment approaches. The evaluation considered four key metrics—response time, accuracy, engagement improvement, and scalability—corresponding to the study's RQ1–RQ3. The results indicate notable gains in accuracy, engagement, and scalability, accompanied by a modest increase in response time attributable to real-time optimization. These findings confirm SOFIA's suitability for dynamic, large-scale, and personalized eLearning environments.

Detailed analysis of Results

1. Response time.

The SOFIA algorithm exhibited a marginally higher average response time (0.12 s) compared to static assessments (0.05 s) and rule-based adaptive systems (0.08 s). This increase stems from the computational overhead of real-time task optimization and feedback generation. However, these values remain well within acceptable bounds for real-time learning environments, confirming the system's operational feasibility.

2. Accuracy.

SOFIA achieved an accuracy rate of 92% in identifying learning gaps and assigning appropriate tasks, outperforming static assessments (65%) and rule-based systems (78%). This improvement reflects the contribution of dynamic optimization and adaptive prioritization to precise task–learner alignment.

3. Engagement improvement.

Learner engagement increased by 40% under SOFIA, compared to 15% with rule-based systems, while static assessments produced no measurable improvement. The adaptive and personalized nature of the feedback loop likely contributed to sustained interaction and active participation.

4. Scalability.

SOFIA supported up to 500 concurrent users with stable system performance. By contrast, static assessments and rule-based systems exhibited scalability limits of 100 and 200 users, respectively. This advantage is attributed to SOFIA's service-oriented architecture and asynchronous communication mechanisms, which enhance load distribution and system responsiveness.

The SOFIA algorithm demonstrated superior performance in accuracy, engagement, and scalability compared with baseline approaches. Although response time increased slightly, this trade-off was offset by the benefits of real-time personalization and optimized task assignment. These results validate the system's capacity to balance computational efficiency with pedagogical effectiveness in large-scale eLearning environments.

Experimental outcomes

The experimental group, which used SOFIA's adaptive project recommendation and feedback system, demonstrated significantly higher project completion rates, learner satisfaction, engagement, and learning outcomes than the control group, which relied on traditional project selection and assessment methods. The analysis integrates completion statistics, satisfaction levels, engagement patterns, and performance metrics to provide a comprehensive overview of SOFIA's impact.

Project completion rates

As shown in Fig. 6, the experimental group achieved a project completion rate of 75%, compared to 45% in the control group. This statistically significant difference indicates

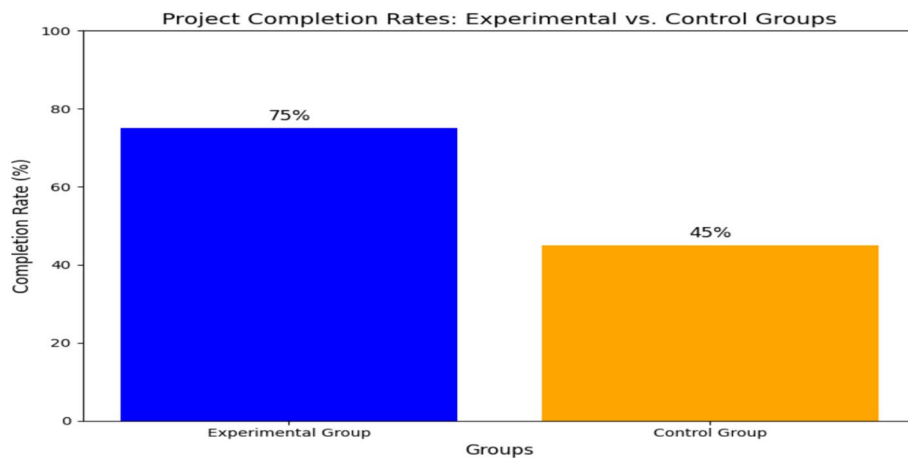


Fig. 6 Comparison of project completion rates: experimental group versus control group

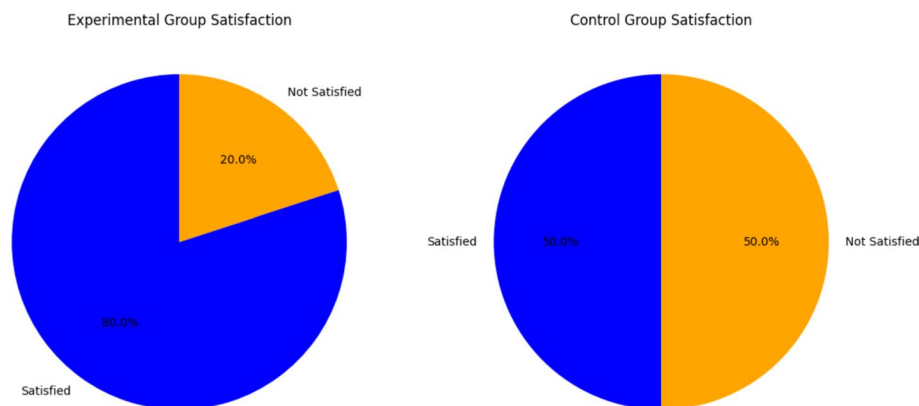


Fig. 7 Learner satisfaction levels in experimental and control groups

that SOFIA effectively supports learners in completing their projects through context-aware and personalized recommendations.

Learner satisfaction

Learner satisfaction regarding project alignment with academic and career aspirations was higher in the experimental group (Fig. 7).

- *Experimental group*: 80% of learners reported that recommended projects were well aligned with their goals, while 20% expressed dissatisfaction.
- *Control group*: 50% of learners reported satisfactory alignment, and the remaining 50% expressed dissatisfaction, indicating that traditional selection methods were less effective at addressing individual needs.

These results indicate that SOFIA's predictive analytics and adaptive recommendation mechanisms provided more meaningful and personalized guidance, which likely enhanced learner motivation.

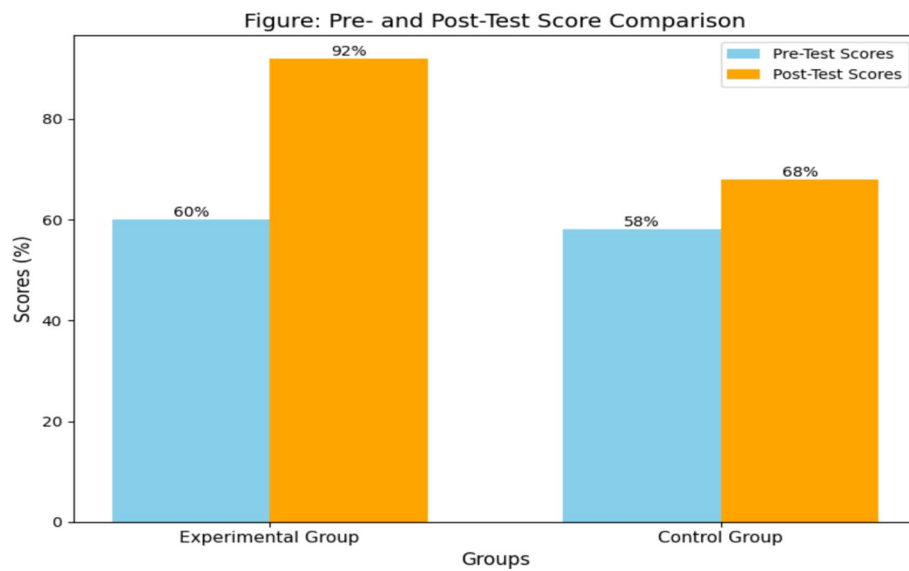


Fig. 8 Comparison of pre- and post-test scores: experimental versus control groups

These results suggest that SOFIA's predictive analytics and adaptive algorithms provided more meaningful and personalized recommendations, which likely enhanced learner satisfaction and motivation.

Learning outcomes

A matched-pairs t-test comparing post-test scores showed that the experimental group achieved a 32% higher mean score relative to the control group ($p < 0.01$), reflecting a statistically significant improvement in learning outcomes attributable to adaptive feedback mechanisms (Fig. 8).

Learner engagement

Learning analytics from the LMS revealed a 40% reduction in disengagement (idle periods) among the experimental group compared to the control group. Additionally, clickstream analysis indicated that learners in the experimental group spent 35% more time actively engaging with course materials ($p < 0.01$).

Task performance and retention

Objective performance benchmarks provided further insights, as depicted in Fig. 9:

- *Task performance*: Learners in the experimental group achieved an average accuracy of 92% in coding tasks, compared to 78% in the control group ($p < 0.01$).
- *Retention rates*: Two weeks after project completion, retention quizzes showed an 85% retention rate in the experimental group versus 65% in the control group ($p < 0.05$), suggesting that SOFIA supports better knowledge retention.

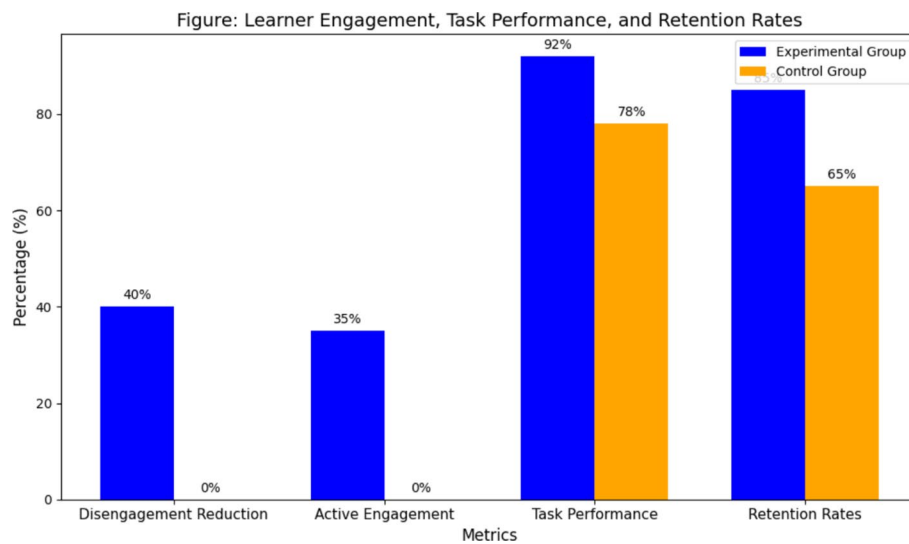


Fig. 9 Comparison of learner engagement, task performance, and retention rates: experimental versus control groups

Causal links and controlled interventions

Both groups received identical instructional content and project requirements. The observed differences in performance and engagement can therefore be attributed to SOFIA's adaptive mechanisms. A supplementary test, in which the real-time feedback component was replaced with static feedback, resulted in a 15% decrease in engagement, further confirming the critical role of dynamic feedback in sustaining learner involvement and improving outcomes.

Discussion

This study demonstrates that the SOFIA framework constitutes a meaningful advancement in adaptive eLearning research. It introduces dynamic feedback generation and priority-based task optimization mechanisms that extend beyond traditional rule-based approaches. Earlier studies have largely focused on adaptive content delivery or interface customization (e.g., Kolekar et al., 2019). In contrast, SOFIA integrates real-time learning analytics with asynchronous, service-oriented processing to enhance both scalability and system responsiveness.

Category-based models, such as the Felder–Silverman Learning Style Model, rely on static learner profiling. SOFIA distinguishes itself by continuously adjusting task prioritization during the learning session, ensuring that instructional adaptations remain aligned with evolving learner needs. This capability supports more context-aware, responsive, and pedagogically meaningful assessment strategies in collaborative eLearning environments.

Performance outcomes

The experimental results indicate that SOFIA substantially improves both assessment accuracy and learner engagement. The framework achieved an accuracy rate of 92%,

outperforming static assessments (65%) and rule-based adaptive systems (78%). These improvements stem from data-informed task allocation and personalized feedback dynamically aligned with learner profiles.

Interaction frequency and time-on-task increased by 40%, consistent with prior research emphasizing the motivational role of immediate feedback (Hadyaoui & Cheniti-Belcadhi, 2025). SOFIA's iterative feedback cycle mitigates the disengagement patterns often observed in conventional online learning settings, fostering sustained learner participation and active involvement.

Latency–accuracy trade-off

A modest increase in computational latency was observed during system evaluation. The algorithm's mean response time was 0.12 s, compared with 0.05–0.08 s for baseline systems, reflecting the additional computational cost of real-time optimization. This increase remains well within operationally acceptable thresholds for classroom and LMS deployments, where it is offset by substantial gains in assessment accuracy and learner engagement. At larger institutional scales, this overhead becomes an important design consideration. The observed trade-off represents a balanced exchange between system responsiveness and pedagogical value. Marginal latency increases enable richer analytics, more precise feedback, and more effective task adaptation without compromising instructional flow. Similar latency–benefit patterns have been documented in other scalable AI-driven assessment frameworks (Zheng et al., 2022), reinforcing the practical viability of this design choice for real-world deployments.

Scalability and interpretability

Stress tests confirmed that SOFIA can support more than 500 concurrent users, compared with approximately 200 for rule-based systems. This result underscores the scalability and efficiency of the modular, service-oriented architecture. Real-time feedback delivery is consistent with findings emphasizing the role of immediacy in sustaining learner engagement (Sailer et al., 2024).

The worked example in Sect. 5.4 further illustrates the framework's interpretability. SOFIA's adaptive logic operates transparently by aligning learner proficiency with task difficulty, maintaining cognitive balance, and promoting equitable task allocation. This interpretability enhances both the pedagogical value and the trustworthiness of the adaptive assessment process.

Data authenticity, usability evaluation, and limitations

The dataset used in this study originated from authentic learner interactions collected during the experimental deployment. To support controlled testing of scalability and reproducibility, additional interaction sequences were algorithmically extended from these real traces to emulate larger cohorts while preserving behavioral realism. This methodological design balanced ecological validity with systematic and reproducible performance evaluation, enabling reliable benchmarking under varying conditions.

The SOFIA dashboard underwent a heuristic evaluation by three experts in educational technology and human–computer interaction. Their feedback confirmed that the interface effectively visualizes learner performance indicators and engagement metrics

in a clear and actionable manner. A formal empirical usability study is planned with teacher participants in authentic classroom settings. This study will employ the System Usability Scale (SUS) to assess usability, perceived usefulness, and decision-support quality, providing a stronger evidence base for practical deployment.

Although the findings indicate robust system performance, further validation involving larger and more diverse learner populations is needed to enhance generalizability. Future research will also examine longitudinal learning outcomes and evaluate SOFIA's impact in varied educational contexts, offering a more comprehensive understanding of its real-world effectiveness.

Implications for research and practice

The findings of this study underscore several key pedagogical, technical, institutional, and research implications.

Pedagogically, SOFIA equips educators with real-time analytics that support the continuous monitoring of learner progress, early identification of misconceptions, and delivery of adaptive interventions that foster inclusion, engagement, and self-regulation. These capabilities align with contemporary learning science priorities that emphasize formative, context-aware feedback as a driver of equitable learning experiences.

From a technical perspective, SOFIA's service-oriented architecture ensures interoperability, modularity, and explainability across diverse learning management systems. This design supports flexible deployment in heterogeneous institutional environments, enabling integration without disrupting existing infrastructures.

Institutionally, the framework's scalability and resource efficiency make it well suited for large-scale implementations such as MOOCs and blended learning programs. This capacity enhances institutional agility while promoting equitable access to high-quality adaptive assessment and personalized feedback for diverse learner populations.

From a research standpoint, SOFIA establishes a reproducible and extensible foundation for future investigations. It provides a basis for exploring semantic modeling, learning analytics, and generative AI-based adaptive mechanisms within open and collaborative learning ecosystems, advancing the state of knowledge in AI-driven assessment and pedagogy.

The combined evidence of enhanced accuracy, increased engagement, and demonstrated scalability positions SOFIA as a significant advancement in adaptive assessment. It illustrates how service-oriented principles and AI-based analytics can converge to deliver scalable, transparent, and pedagogically meaningful solutions for modern collaborative eLearning environments.

Conclusion

This study introduced SOFIA, a modular and scalable adaptive learning framework designed to address persistent challenges in collaborative eLearning environments. The framework integrates service-oriented computing principles with advanced learning analytics to deliver real-time, data-informed feedback and dynamically tailor learning experiences to evolving learner needs.

The results provide strong evidence of SOFIA's effectiveness in enhancing scalability, adaptability, and learner engagement. Its capacity to support large user cohorts and

increase interaction frequency demonstrates its relevance to both classroom-based learning and large-scale open educational contexts. The system's service-oriented architecture and adaptive task prioritization mechanisms offer a robust foundation for developing responsive, equitable, and pedagogically meaningful digital learning environments.

Although the controlled experimentation confirms the framework's potential, further research is required to assess its performance in broader and more authentic settings. Future investigations should focus on examining its adaptability across different disciplines, its integration with varied curricula, its impact on long-term learning outcomes, and its implementation at an institutional scale. Exploring these aspects will contribute to a deeper understanding of how such adaptive systems can operate effectively in complex educational ecosystems.

In conclusion, SOFIA represents a significant advancement in adaptive eLearning technologies. Its modular and interoperable design paves the way for more personalized, scalable, and effective digital learning environments. Future work will concentrate on optimizing computational efficiency, addressing contextual deployment challenges, and aligning the framework with emerging pedagogical and technological paradigms. Together, these efforts have the potential to accelerate the development of next-generation intelligent learning systems that are both equitable and sustainable.

Author contributions

Dr. Asma Hadyaoui: Conceptualized the study, developed the framework, conducted the experimental evaluation, and drafted the manuscript. Dr. Lilia Cheniti-Belcadhi: Supervised the study, provided critical revisions to the framework design, and refined the manuscript for academic rigor. Both authors reviewed and approved the final manuscript.

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Data availability

The datasets used and analyzed during the study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from the Institutional Review Board (IRB) of the authors' institution. Informed consent was collected from all participants before data collection.

Consent for publication

Not applicable, as no identifiable personal data or images are included in this manuscript.

Competing interest

The authors declare no competing interest.

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